F78PA Examination December 2019

Feedback on class performance (Edinburgh)

General comments:

In general, performance in Edinburgh was not at all strong and some scaling of raw marks and adjustment to grade boundaries was made by the examination board. This was justified on the grounds that the exam was felt to be rather long, and that, due to a typo in question 5, this question may have caused some students confusion (although students were not penalised for answering question according to typo). On the positive side, there were some very good performances and some students had clearly mastered the important ideas in what is quite a challenging subject.

Some specific comments on performance:

A number of general weaknesses were apparent. These related to the following issues.

1. Some students struggled with basic mathematical skills such as sketching graphs of functions correctly (Q1 (a), Q3 (a)) or manipulating logarithms effectively (Q 6(d)). These are very important in all mathematical courses.
2. Some students were very unclear on important definitions and, other aspects of bookwork, such as:
    - Marginal and conditional distributions (Q4(a));
    - Covariance and correlation, and important properties thereof (Q7(a));
    - Statement of inequalities such as Chebyshev's inequality (Q6(e));
    - Central limit theorem and its use (Q2(b));
    - Properties of expectations of linear functions of random variables (Q2(b)).
    - General procedures, such as calculations of expectations of functions or random variables (Q1(d), Q2(b), Q2(c)), or procedures for computing pdfs for transformations of continuous random variables (e.g Q3(b)).

    These general definitions and procedures will be very used many times in later statistics and probability courses including F78PB so it is important that you have a clear understanding of them.
3. **Other mathematical skills:** In some questions such as Q2(b) or Q4(c), students failed to spot the quickest way to solve the question and sometimes carried out lengthy calculations that could have been avoided, using up valuable time. Further experience of studying mathematics courses will help to build skills of this nature. In general, it is always worth pausing to reflect on the most efficient way to solve a problem before embarking on the solution.

Going forward:

If you did not perform well in F78PA you will nevertheless have the chance to reinforce its learning outcomes in F78PB. It is therefore very important that you attend lectures and tutorials and fully engage with that course. Lecture and tutorial attendance was in general poor for F78PA and a number of students missed out on key information as a result.

Revision sessions:

I shall be happy to assist students who are required to resit with their revision through 1-1 sessions or via revision lectures which I intend to organise closer to the August resit. In the meantime, please don't hesitate to contact me for help or advice.